

SMART INDIA HACKATHON 2026



SAT-SA: Security Audit & Supervisory Analytics

Turning SOC Records into Actionable Audit Evidence

Problem Statement ID – SIH26157

Problem Statement Title – Supervisory Analytics Tool for SOC Assessment

Theme – Blockchain & Cybersecurity

PS Category – Software (Offline / Air-Gapped)

Team ID – 144037

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SOLUTION SCOPE & HUMAN-IN-THE-LOOP AUDIT OBJECTIVE

An offline supervisory analytics tool that evaluates SOC records, detects behavioural anomalies, execution gaps, and missing evidence, and prioritises high-risk entities to assist human auditors.

Offline-First Supervisory Analytics & Review Prioritisation for Critical Sector Security Operations Centers
Human-in-the-loop decision support enabling defensible, evidence-backed supervisory investigations.

1. THE PROBLEM

High Data Volume

SOCs generate millions of alerts, cases, closures, and asset records.

Limited Expert Review Capacity

Large SOC datasets make exhaustive manual review difficult.

Missing Evidence

Silent critical assets & dropped logs produce zero events to sample.

2. OUR PROPOSED SOLUTION

SOC Records

Historical Alerts, Cases, Assets



SAT-SA Analytics Engine

• Rule Analysis • Statistics (EB)
• Peer Cohorts • Anomaly ML



Review Priorities

Prioritised Entities & Findings



Human Expert Auditor

Targeted Forensic Investigation

3. INNOVATION & UNIQUENESS

01 — EXECUTION GAPS

Evidence exists, but behaviour appears unusual (e.g. 4s fast closures, template notes).

02 — NEGATIVE SPACE

Expected evidence is missing or unexpectedly sparse (e.g. silent SCADA assets, blackout shifts).

03 — EVIDENCE TRAIL

Every recommendation links to an unbroken chain of verifiable raw records.

**"SAT-SA analyses not only what exists — but also what should exist."
Identifies where expert review should focus without replacing the human investigator.**

TECHNICAL APPROACH

Methodology, Workflow & Detection Engine

COMPLETE SYSTEM WORKFLOW PIPELINE

DATA INPUT

NORMALISATION

FEATURE EXTRACTION

DETECTION ENGINE

PRIORITISATION

EVIDENCE TRAIL

HUMAN REVIEW

DATA INPUT & DETECTION ENGINE

- **Ingestion Sources:** CSV | JSON | DB Exports (Alerts, Cases, Closures, Assets, Telemetry)
- **4 Parallel Analytics Branches:** Rules • Statistical Analysis (EB) • Peer Benchmarks • Anomaly ML
- **Dual-Paradigm Output:** Splits into Execution Gaps (Faulty actions) + Negative Space (Omissions)

EXECUTION GAP VISUAL (Alert A-19283)

Opened: 10:02:15 → Closed: 10:02:19 | Duration: 4 seconds

Significant Peer Deviation → [Review Recommended]

NEGATIVE SPACE VISUAL (Critical SCADA Gateways)

Expected: ██████████ (Baseline Activity)

Observed: ████████ (Low Telemetry) → [Potential Coverage Gap]

TECHNOLOGY STACK & OFFLINE-FIRST ARCHITECTURE

Python 3 • Pandas • DuckDB + Parquet • scikit-learn • Streamlit / Plotly • Pydantic

OFFLINE-FIRST: Air-Gapped • No External Cloud APIs • Tamper-Evident SHA-256 Ledger

Designed for high-volume local analytical processing with zero external cloud API dependencies.

FEASIBILITY AND VIABILITY

Feasibility Analysis & Risk-Mitigation Matrix

FEASIBILITY ANALYSIS

OFFLINE DEPLOYMENT

Works completely without cloud access, external APIs, or outbound network calls in air-gapped environments.

SCALABLE ANALYTICS

Columnar storage (DuckDB/Parquet) and vectorized processing easily handle large operational datasets locally.

MODULAR DETECTORS

Detector rules, statistics, and anomaly models can be independently updated and verified.

HUMAN-IN-THE-LOOP

System prioritises and recommends review; human experts retain absolute assessment authority.

CHALLENGES & SAT-SA MITIGATION STRATEGIES

Potential Challenge	SAT-SA Mitigation Strategy
Incomplete / Missing data	Automated data-quality gates & schema validation
Different CSE data schemas	Canonical Pydantic schema maps heterogeneous formats
Small sample sizes per entity	Empirical Bayes Poisson-Gamma statistical shrinkage
False alarm fatigue	Context-aware peer benchmarking + human review
Adversarial KPI gaming	Multi-detector diversity + random audit baseline

OFFLINE DEPLOYMENT WORKFLOW

Offline Dataset → SAT-SA Engine → Local DuckDB → Local Dashboard → Expert Review

✓ Zero dependency on external cloud services or Internet connectivity.

Designed for air-gapped deployment and suitable for further validation in supervisory workflows.



IMPACT AND BENEFITS

Supervisory Transformation & Strategic Value

SUPERVISORY TRANSFORMATION

TODAY: MANUAL DATA OVERLOAD

Large SOC Dataset → Manual Sampling → Limited Review Capacity

▼ SAT-SA ANALYTICS ▼

WITH SAT-SA: EVIDENCE-DRIVEN REVIEW

Evidence-Based Priorities → Explainable Findings → Focused Expert Review
Evidence-backed and traceable supervisory assessment

FOUR PILLARS OF SUPERVISORY VALUE

FOCUSED REVIEW

Helps experts identify where to investigate first.

EVIDENCE-DRIVEN

Links recommendations directly to underlying records.

BETTER VISIBILITY

Detects both unusual behaviour and missing evidence.

EFFICIENT AUDIT

Reduces dependence on purely manual sampling.

TRADITIONAL MANUAL REVIEW VS SAT-SA

Traditional Manual Review	SAT-SA Supervisory Analytics
Sample-driven manual review	Priority-driven analysis across the submitted dataset
Record-focused in isolation	Behaviour-focused across time
Hard to compare peer entities	Context-aware peer benchmarking
Inspects existing logs only	Inspects existing + missing evidence

"From Data Overload → Evidence-Driven Review"

SAT-SA helps supervisory experts spend limited review time where evidence indicates the greatest need.

Regulatory Guidelines, Standards & Scientific References

1. NCIIPC & GOVERNMENT GUIDELINES

- NCIIPC Guidelines (Sec 70A, IT Act 2000) — [Official NCIIPC](#)
- CERT-In Cyber Security Directions — [Official CERT-In](#)
- NCIIPC / QCI Conformity Assessment Framework — [Official NCIIPC Guidelines](#)

2. SMART INDIA HACKATHON 2026 SPECIFICATION

- Problem Statement 26157: Supervisory Analytics Tool — [Official SIH](#)
- Objective: Offline supervisory audit engine for prioritizing human investigation
- Scope: Analysis of historical alerts, case tickets, shift rosters, and asset telemetry

3. RESEARCH ON SOC ANALYTICS & OPERATIONS

- MITRE ATT&CK Enterprise Framework & Detection Taxonomy — [Official MITRE](#)
- SOC Alert Fatigue & Supervisory Triage Optimization (IEEE/ACM Surveys)
- Empirical Analysis of SOC Workflows: Closure Velocity, Shift Drift & Quality

4. STATISTICAL & MACHINE LEARNING REFERENCES

- Isolation Forest (Liu, Ting & Zhou) — [IEEE ICDM](#)
- MinHash / LSH (Broder et al.) — [IEEE FOCS](#)
- Empirical Bayes Poisson-Gamma Shrinkage (Efron & Morris) — [JASA](#)

All methodologies grounded in established supervisory standards, peer-reviewed statistical models, and official regulatory guidelines.